Joonhyuk Ko

Curriculum Vitae

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Education

Princeton University — Princeton, NJ
Ph.D. in Computer Science — *Aug 2026 – May 2031*

University of Virginia — Charlottesville, VA
B.S. in Computer Science, B.A. in Mathematics — *Aug 2022 – May 2026*
GPA: 3.997/4.000; Highest Distinction

Research Experience

Princeton University, Department of Computer Science — Princeton, NJ
Graduate Research Assistant — *Aug 2026 – Present*
Advisor: **Prof. Lydia T. Liu**

University of Virginia, Department of Computer Science — Charlottesville, VA
Undergraduate Research Assistant — *Jan 2024 – Jul 2026*
Advisor: **Prof. Ferdinando Fioretto**

Carnegie Mellon University, School of Computer Science — Pittsburgh, PA
Summer Research Intern — *May 2025 – Aug 2025*
Advisor: **Prof. Zhiwei Steven Wu**

Publications

$(\alpha - \beta)$ *denotes alphabetical author ordering.*

Conference Proceedings

C2. $(\alpha - \beta)$ Andrew Ilyas, **Joonhyuk Ko**, Jingwu Tang, Steven Wu, Jiahao Zhang. *"Provably Label-Efficient Conformal Prediction."*
International Conference on Machine Learning (**ICML**), 2026.

C1. **Joonhyuk Ko**, Juba Ziani, Saswat Das, Matt Williams, Ferdinando Fioretto. *"Fairness Issues and Mitigations in (Differentially Private) Socio-Demographic Data Processes."*
AAAI Conference on Artificial Intelligence (**AAAI**), 2025. **Oral Presentation (Top** 5%**).**

Preprints

P1. **Joonhyuk Ko**, Juba Ziani, Ferdinando Fioretto. *"Optimal Allocation of Privacy Budget on Hierarchical Data Release."*
ArXiv Preprint, 2025.

Awards and Honors

NSF Graduate Research Fellowship (NSF GRFP) [link] [news] — 2026
Five-year fellowship in STEM with three years of financial support (total award: $159,000).

CRA Outstanding Undergraduate Researcher Award [link] [news] — 2025, 2026
Finalist (2026), Honorable Mention (2025).
Recognizes North American undergraduates with outstanding computing research potential.

Louis T. Rader Undergraduate Research Award [link] — 2025
Top Honors. The highest undergraduate CS research award at the University of Virginia ($750 award).

The Raven Society Member — 2025
The oldest and most prestigious honorary society at the University of Virginia.

AAAI 2025 Student Travel Award 2025
Received NSF-sponsored travel funding to support conference participation ($1,800 award).

Dean's Undergraduate Engineering Research Fellowship 2024
Received a stipend to pursue full-time research for summer 2024 ($4,800 award).

Invited Talks

Talk: *Fairness Issues and Mitigations in (Differentially Private) Socio-Demographic Data Processes.*
TOC4Fairness Seminar, Simons Foundation. [link] *Mar 2025*

Teaching Experience

Teaching Assistant
APMA 6410 (Graduate-level): Engineering Mathematics I (Fall 2023)
APMA 2130: Ordinary Differential Equations (Spring 2023)

Additional Information

Technical Skills: Python, C, MATLAB, LaTeX, SLURM, Gurobi, CVXPY, SciPy, TypeScript
Languages: English, Korean
Nationality: U.S. Citizen